

CALORIE TRACKER & FOOD LOG

A Simple Python Console Program to Count Daily Food Calories

About This Project:

This report presents a simple Python mini-project built for everyday calorie counting. The program stores nutritional values for common foods—especially everyday Indian home-cooked meals like roti, dal, rice, and sabzi. Users can type in what they ate and how many servings they had, and the program calculates their total calories.

Submitted By:	Sameer Singh
Registration Number:	26BAI10004
Course:	BTech CSE AI-ML
Subject:	Introduction to Programming and Problem Solving
Semester:	Fall Semester 2026-27

1. INTRODUCTION

Counting calories is one of the easiest ways to stay healthy, but most people stop doing it because mobile apps are too complicated. Big apps require you to make an account, deal with constant ads, stay connected to the internet, and search through thousands of foreign foods just to log a simple meal.

To solve this, I wrote a small, straightforward Python script that runs right in the terminal. It works completely offline, needs no setup or third-party libraries, and focuses on simple, fast calorie calculation.

2. PROBLEM STATEMENT

Most calorie apps are made with Western diets in mind. When you try to log normal Indian meals—like 2 rotis, a bowl of dal tadka, or a plate of poha—it is hard to find accurate or straightforward numbers. People usually have to guess or give up.

The goal of this project is to create an easy console program that stores common Indian meals with pre-set calorie values. The user should be able to type what they ate, enter their serving size, and immediately get the total calories.

3. WHAT THE PROGRAM DOES (FUNCTIONAL REQUIREMENTS)

The program was designed to carry out these main tasks:

- **Display food catalog:** When the program starts, it prints out the complete list of available items and their calorie values so the user knows what they can pick.
- **Food search & match:** The user enters the name of a food. The program ignores accidental spaces and matches uppercase or lowercase letters smoothly.
- **Serving calculation:** The user enters the quantity (including decimals like 0.5 or 1.5). The program multiplies the serving count by the item's calories.
- **Running updates:** After each food item is entered, the program immediately confirms how many calories were added.
- **Final summary:** When the user types done, the program prints a clean list of everything logged and the total calorie count.

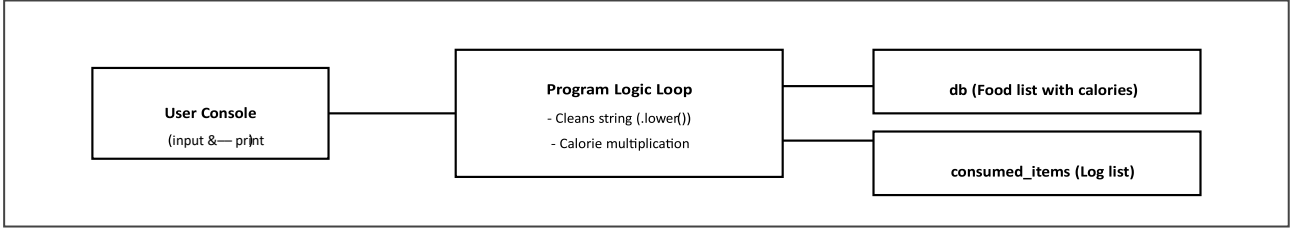
4. GENERAL SYSTEM REQUIREMENTS (NON-FUNCTIONAL)

Along with the main features, the project follows these practical rules:

- **Simple & clean interface:** Clear command-line prompts guide the user step by step.
- **No external libraries:** Written entirely in pure Python standard library. Anyone with Python installed can run it instantly.
- **Basic error safety:** If the user enters a food that is not in the list, the program does not crash; it shows a helpful error and lets them try again.

5. SYSTEM ARCHITECTURE

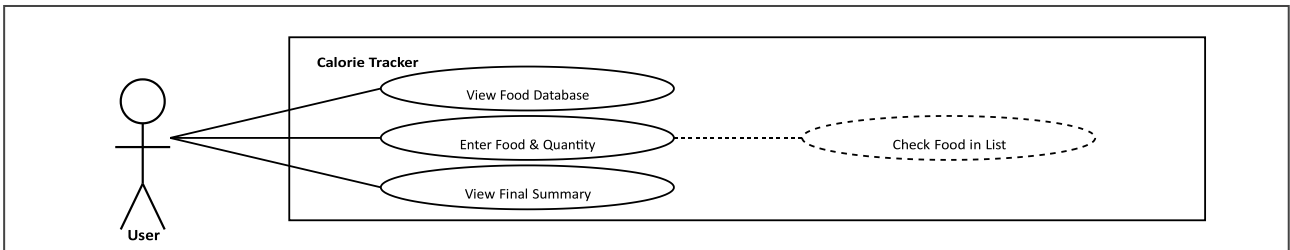
The program is split into three simple parts: User Input/Output (the terminal screen), Program Logic (searching the list and doing the math), and In-Memory Data (the list of foods and the logged items).



6. DESIGN DIAGRAMS

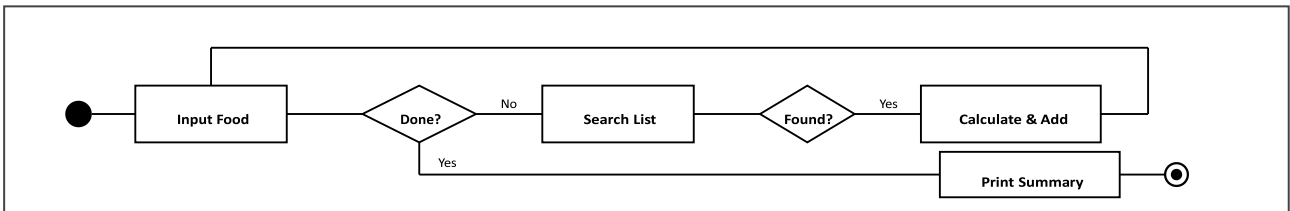
6.1 Use Case Diagram

This diagram shows the actions the user can perform with the script:



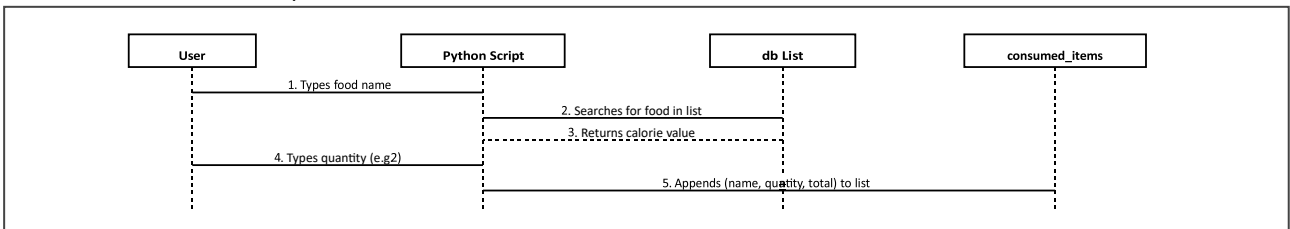
6.2 Workflow (Step-by-Step) Diagram

This flowchart shows how the program runs from start to finish:



6.3 Sequence Diagram

This shows the order of steps when a user enters an item:



6.4 Data Structures & Storage

Instead of a complex external SQL database, the project keeps things simple by using Python's built-in data types:

- **Food Database (db):** A Python list containing tuples: ("apple", 95.0). Tuples are ideal here because food calorie values are fixed and shouldn't accidentally change.
- **Food Log (consumed_items):** A dynamic list where each logged meal is saved as (name, quantity, item_total).

7. DESIGN DECISIONS & CHOICES

- **Why a Python list of tuples?** A small list of 36 items is extremely fast to search and avoids having to set up SQL files or external database drivers.
- **Why .strip().lower()? If someone types " Apple " or "APPLE",** the program trims the extra spaces and changes it to lowercase so it matches "apple" without errors.
- **Why use float for quantities?** People often eat half a bowl or two and a half rotis. Using floats makes entries like 0.5 or 1.5 work properly.

8. IMPLEMENTATION & CODE DETAILS

The whole project is written in a single file called `calorie_calculator.py`. The program starts by printing the database with a for loop. Then it enters a while True loop that repeatedly asks for food names. If the user enters "done", the loop breaks using the break statement and prints the final bill.

9. PROGRAM OUTPUT / SCREENSHOTS

Here is the actual output from running the program in the terminal:

```

- khichdi (1 cup) (180.0 cal)
- idli (1 piece) (58.0 cal)
- plain dosa (1 piece) (133.0 cal)
- masala dosa (1 piece) (250.0 cal)
- sambar (1 bowl) (90.0 cal)
- medu vada (1 piece) (145.0 cal)
- upma (1 cup) (190.0 cal)
- poha (1 cup) (180.0 cal)
- samosa (1 piece) (262.0 cal)
- pakora (1 piece) (60.0 cal)
- cucumber raita (1 bowl) (75.0 cal)
- gulab jamun (1 piece) (150.0 cal)
- masala chai with milk & sugar (1 cup) (80.0 cal)
- sweet lassi (1 glass) (210.0 cal)
_____X_____X_____X_____X_____X_____
Enter 'done' when you finish logging food.

Enter food item: done
_____X_____X_____SUMMARY_____X_____X_____
No items were logged.
_____X_____X_____X_____X_____X_____

problems  Output  Debug Console  Terminal  Ports
- upma (1 cup) (190.0 cal)
- poha (1 cup) (180.0 cal)
- samosa (1 piece) (262.0 cal)
- pakora (1 piece) (60.0 cal)
- cucumber raita (1 bowl) (75.0 cal)
- gulab jamun (1 piece) (150.0 cal)
- masala chai with milk & sugar (1 cup) (80.0 cal)
- sweet lassi (1 glass) (210.0 cal)
_____X_____X_____X_____X_____X_____
Enter 'done' when you finish logging food.

Enter food item: rice cup
Enter quantity/servings of rice cup: 2
Added: 412.0 calories.

Enter food item: apple
Enter quantity/servings of apple: 2
Added: 190.0 calories.

Enter food item: done
_____X_____X_____SUMMARY_____X_____X_____
rice cup x 2.0 = 412.0 calories
apple x 2.0 = 190.0 calories
_____X_____X_____X_____X_____X_____
Total Calories Consumed: 602.0 kcal
_____X_____X_____X_____X_____X_____

- pakora (1 piece) (60.0 cal)
- cucumber raita (1 bowl) (75.0 cal)
- gulab jamun (1 piece) (150.0 cal)
- masala chai with milk & sugar (1 cup) (80.0 cal)
- sweet lassi (1 glass) (210.0 cal)
_____X_____X_____X_____X_____X_____
Enter 'done' when you finish logging food.

Enter food item: kachori
Error: 'kachori' not found in database. Try again.

Enter food item: 
```

10. TESTING APPROACH

I tested the program manually by trying different normal and abnormal inputs:

- **Normal entry:** Entered apple with quantity 2. Result: correctly printed 190.0 calories.
- **Food not in database:** Entered kachori. Result: displayed an error message and let the user try again without crashing.
- **Exiting immediately:** Typed done at the first prompt. Result: neatly printed "No items were logged."

11. CHALLENGES FACED

- **Food Names:** For foods like roti or dal, calories depend a lot on size. I solved this by adding portions directly into the names, like roti (1 medium) or dal tadka (1 bowl).
- **Handling the exit condition:** Initially, typing done caused the code to search for "done" in the food database and print an error. I fixed this by checking if name == "done": break before starting the database loop.

12. LEARNINGS, FUTURE WORK & REFERENCES

Learnings: Building this project helped me get comfortable with core Python concepts like lists, tuples, loops (while and for), string methods (.strip() and .lower()), and writing clean terminal prompts.

Future Enhancements: In the future, I plan to let users add their own custom foods during runtime and save their daily summaries to a text file so their history isn't lost when the program closes.

References: Python Official Documentation (python.org) & National Institute of Nutrition (ICMR-NIN) Indian Food Composition Reference Tables.